

RESEARCH METHODS

DAY 5

Welcome to Day 5!

Let's Reflect on our Process

What are your thoughts?

What are the challenges you are facing?

What can be done next?

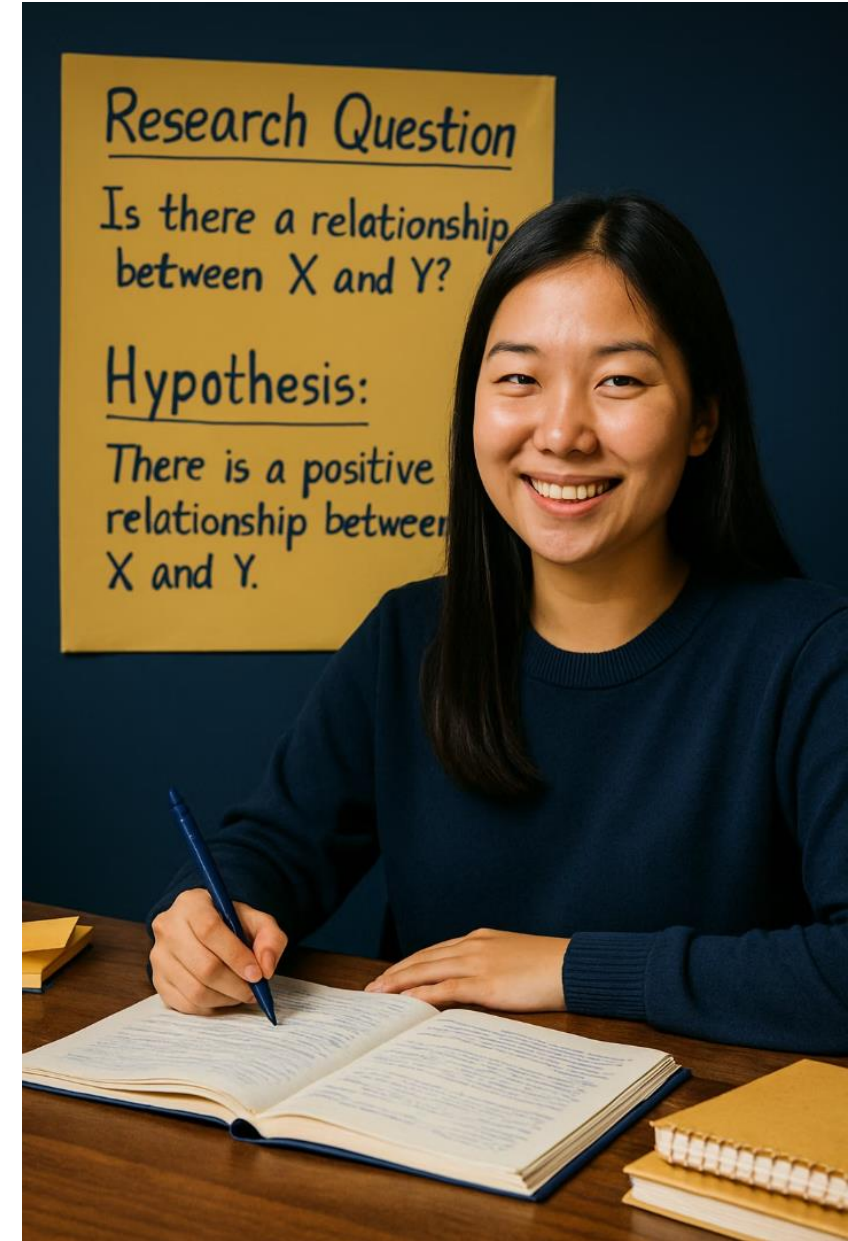


REFINEMENT IS NORMAL!

Questions and hypotheses often evolve as you learn more.

But how do you “learn more”?

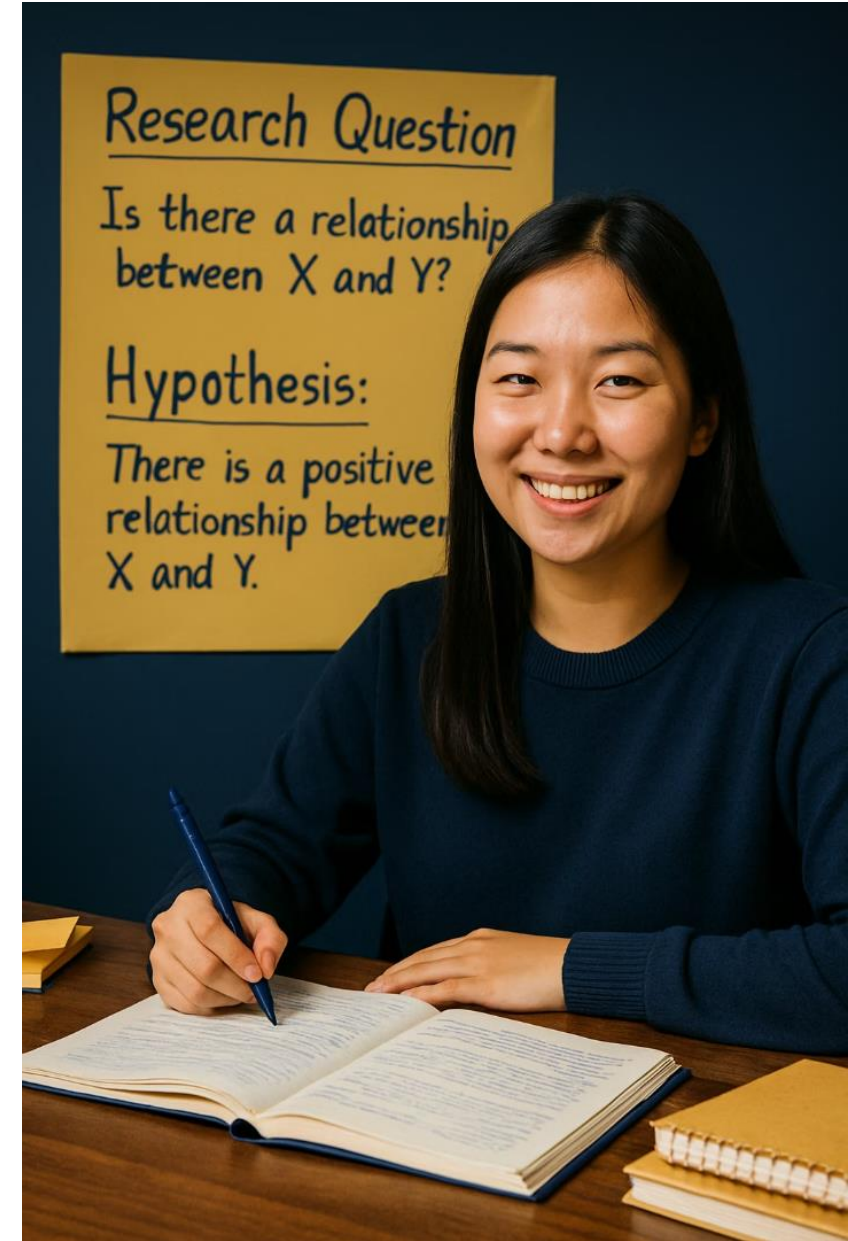
Answer: **Literature Review**
Next phase of research!



SMALL PROGRESS IS STILL PROGRESS

You are just beginning this
journey.

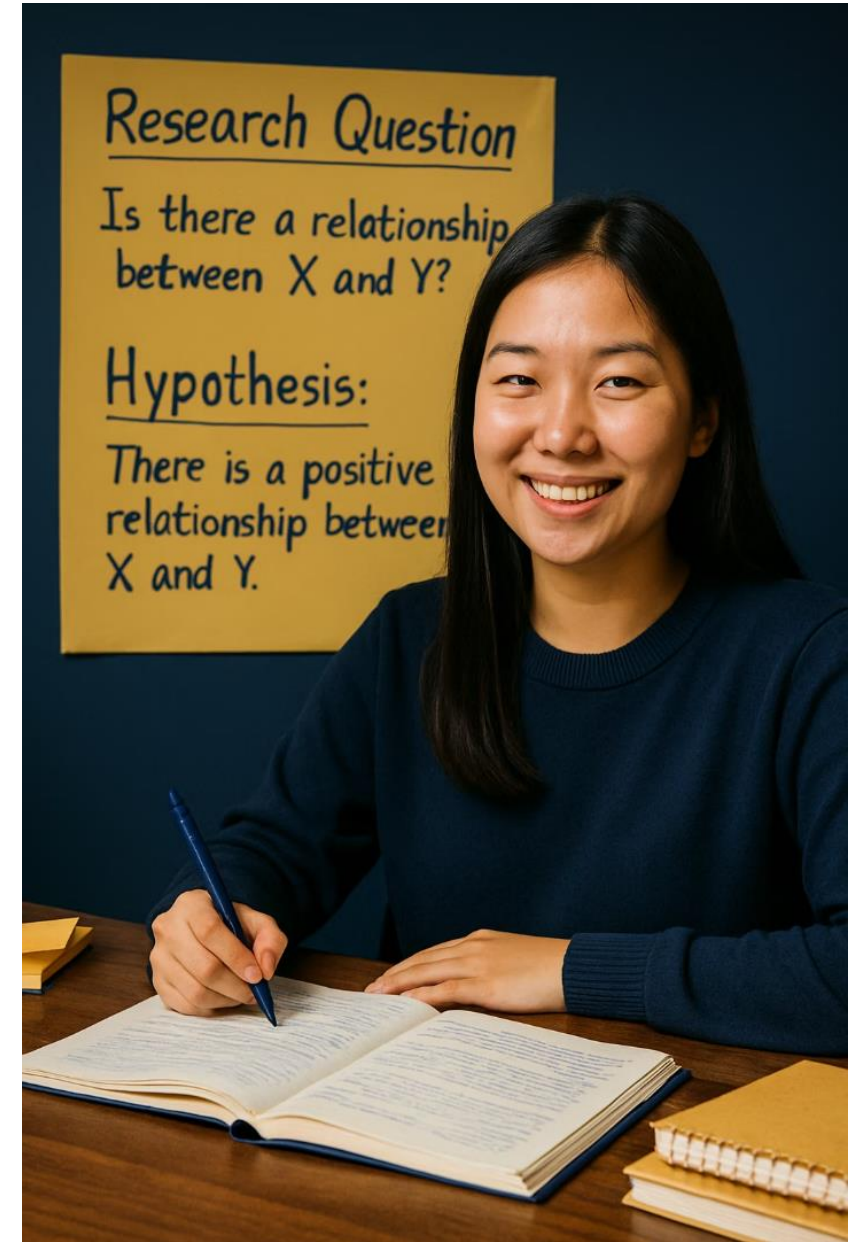
Allow yourself to take smaller
steps. And **trust the process.**



HOW YOU LOOK AT IT IS A GAME-CHANGER

If you see it like “spending your time reading articles” vs. making an exploration.

Take 2-3 key points from a resource and keep on with exploring.



Now, Let's Continue with Planning

Reliability & Validity

Sampling



Reliability and Validity

Reliability

Consistency of results if repeated under the same conditions.

- Would you get similar results if repeated?
- Are measurements consistent?
- Is your procedure standardized?



Improving Reliability



Standardize Procedures

Use the same methods, instructions, and conditions for all participants or trials.



Multiple Trials

Repeat measurements or tests to see if results are consistent.



Check Instruments

Ensure measuring tools are working properly and giving consistent readings.



Larger Sample Size

Include more participants or items to reduce the impact of individual variations.

Reliability and Validity

Reliability

Consistency of results if repeated under the same conditions.

Validity

Whether you're actually measuring **what you intend to** measure.

- Does your method truly address your question?
- Are your measurements relevant?
- Can you draw appropriate conclusions?



Good research needs both reliability and validity to be trustworthy.

Improving Validity

Align Methods with Questions

Choose research approaches that directly address what you want to know.

Use Multiple Measures

Collect different types of data to get a more complete picture of what you're studying.

Control Variables

Identify and minimize factors that could confuse your results.

Pilot Test

Try out your methods on a small scale first to identify and fix problems.



Aisha's Research Revisited

Initial Approach (Poor)

- Different music for each friend
- "Productivity" measured by self-reports of "fine" or "good"
- Different tasks for each person
- Inconsistent reporting methods

Improved Approach (Better)

- Same calm music playlist for all
- Productivity measured by task completion and quiz scores
- Same reading and math tasks for everyone
- Consistent 30-minute sessions



The improved design has better reliability and validity, leading to more trustworthy results.

Sampling: Who or What Will You Study?



*Good sampling makes your
results more reliable,
generalizable and valid.*

Sampling: Who or What Will You Study?

What is a Sample?

A smaller subset of the population or items you're interested in.

Since studying everyone or everything is usually impossible, researchers select **representatives**.

Key Sampling Questions

- How many people/items will you include?
- How will you select them?
- Will your sample represent the larger group?
- What potential biases might affect your selection?

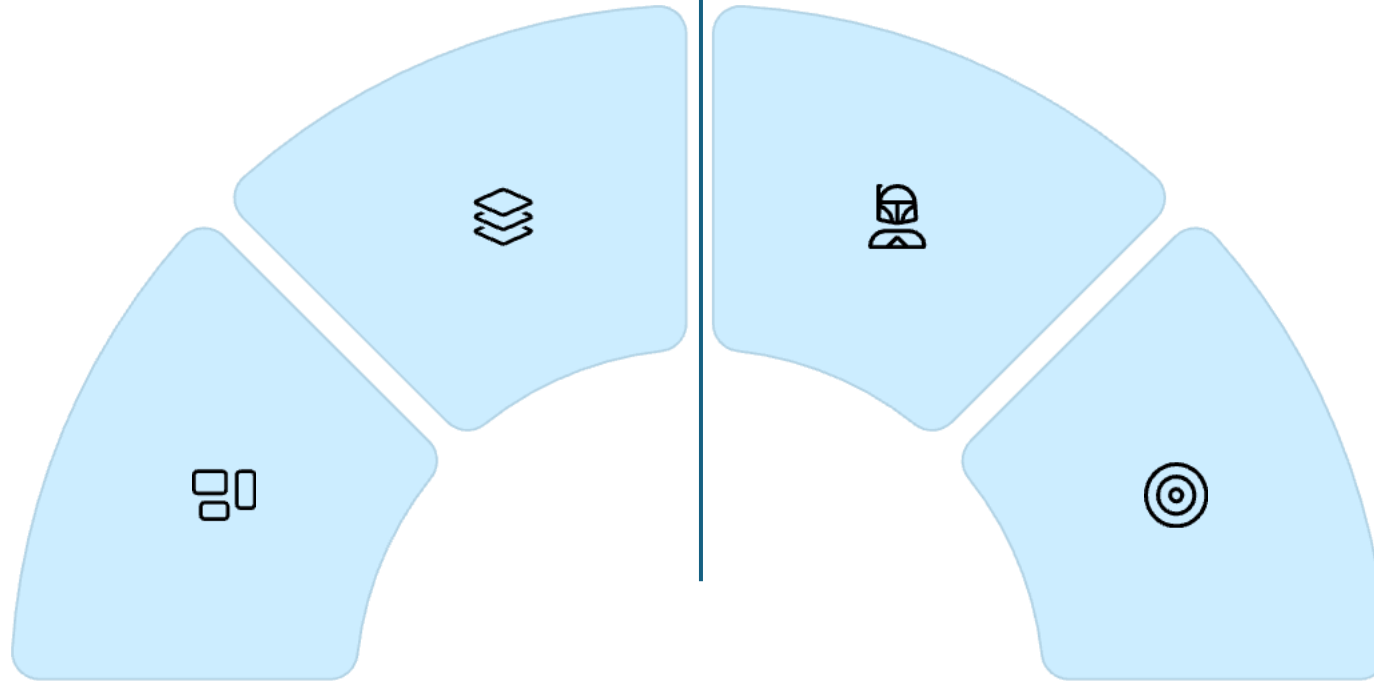


Good sampling makes your results more reliable, generalizable and valid.

Types of Sampling

Random Sampling

Everyone has equal chance of selection (like drawing names from a hat).



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Random Sampling

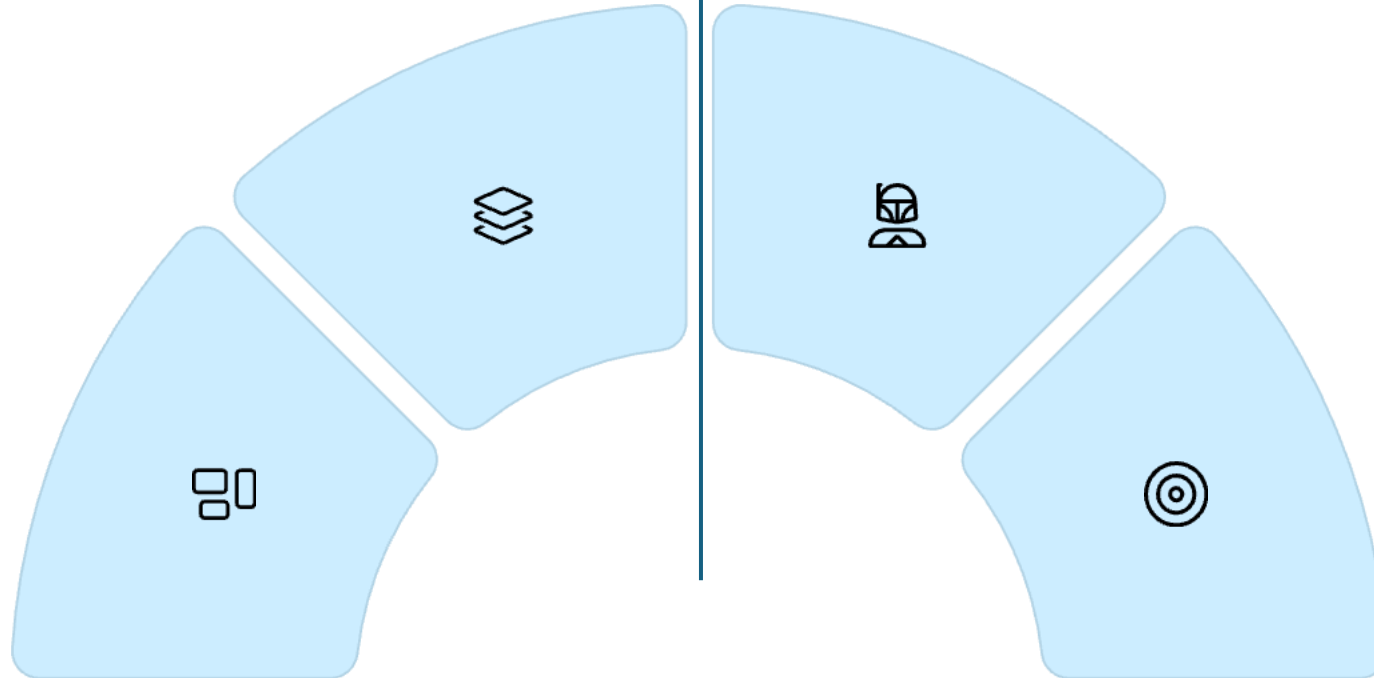
Everyone has equal chance of selection (like drawing names from a hat).

Stratified Sampling

Select from important subgroups to ensure representation (e.g., different grades).

Cluster Sampling

Randomly select entire groups (randomly choose 3 classrooms)



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Stratified Sampling

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Convenience Sampling

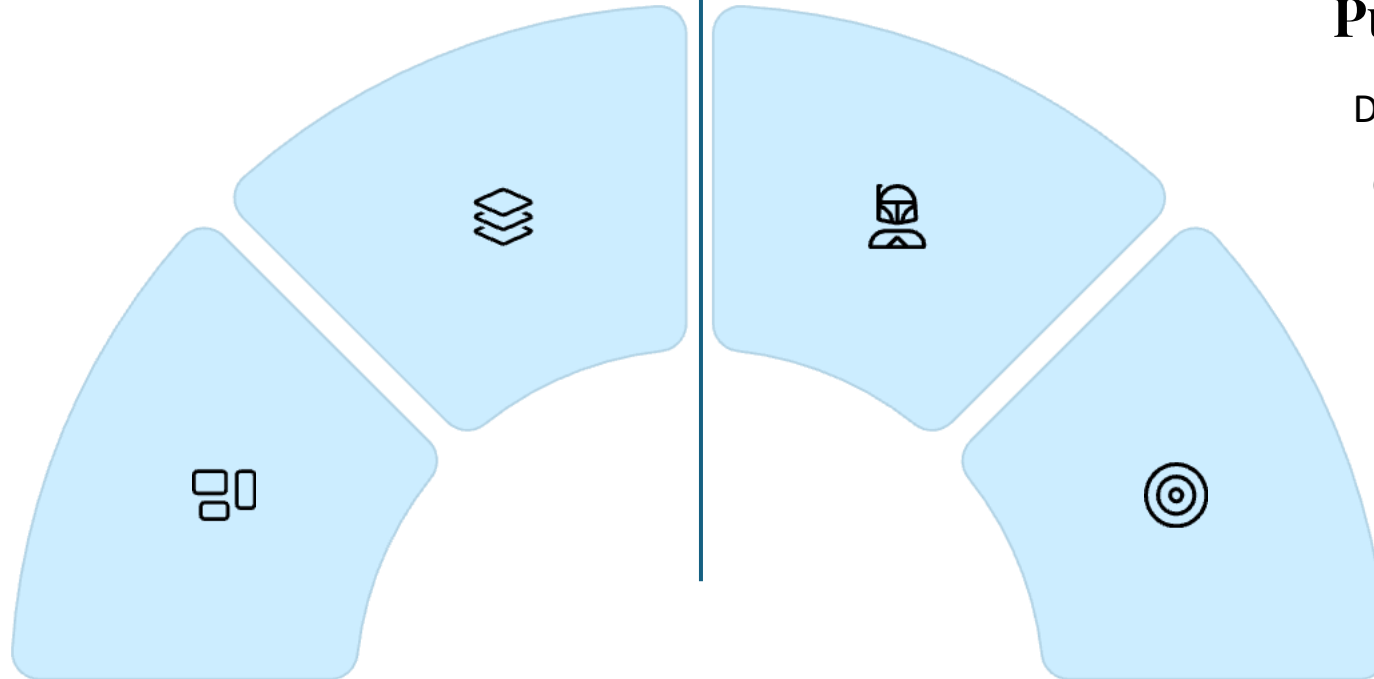
Select whoever is easiest to reach (like your own class).

Purposive Sampling

Deliberately select specific cases based on research needs.

Cluster Sampling

Randomly select entire groups (randomly choose 3 classrooms)



Sampling Examples

Survey

Instead of surveying all 1,200 students in a school, randomly select 120 students across all grade levels.

Plant Experiment

Use 30 plants of the same species and size, randomly assigning 15 to experimental and 15 to control groups.

Observation

Instead of observing a park 24/7, sample time periods by observing for 1 hour on mornings and 1 hour on afternoons.

Historical Research

Instead of reading every newspaper from 100 years, sample by examining the first issue of each month from selected years.



Sampling Bias

What is Sampling Bias?

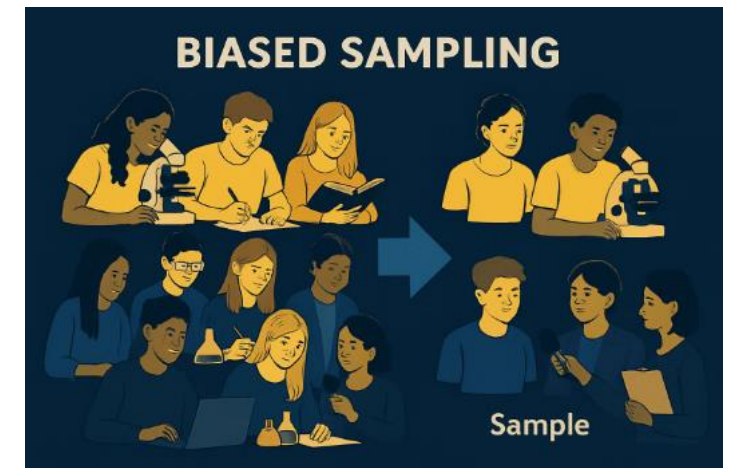
When your sample doesn't fairly represent the population you're studying.

This can skew results and lead to incorrect conclusions.

Example

Surveying only cooking club members about school lunch preferences would give biased results.

These students likely have different food interests than the general student body.



Even with limited resources, try to broaden your sample to reduce obvious biases.



Practical Sampling Tips



Broaden Your Reach

Instead of just surveying friends, include students from different classes or groups.



Consider Balance

Try to include representation from important subgroups (different grades, genders, etc.).



Use Random Selection

When possible, use methods like drawing names or random number generators to select participants.



Acknowledge Limitations

Be transparent about who your sample represents and any potential biases.

Materials and Resources

▷◁ Equipment

🔑 Access & Permissions

\$ Budget

▷◁ People





Equipment Planning



Technology

Computers, tablets, cameras, recording devices, or specialized equipment.



Documentation Tools

Survey forms, observation checklists, interview guides, or data collection sheets.



Lab Supplies

Chemicals, test tubes, measuring tools, or experiment-specific materials.



Measurement Instruments

Stopwatches, rulers, scales, thermometers, or other measuring devices.

Budget Considerations

Low-Cost Alternatives

Look for free or inexpensive substitutes for costly materials. Borrow equipment when possible.

School Resources

Check what supplies your school already has available for student projects.

Funding Options

Consider mini-grants from school clubs, PTAs, or community organizations for special projects.

Scale Appropriately

Adjust your project scope to match available resources rather than overextending.



Access and Permissions



Physical Spaces

Permission to use classrooms, labs, libraries, or community spaces for your research.



Participant Consent

Permission from people (and parents for minors) to participate in your study.



Information Access

Approval to use school records, databases, or other information sources.



Institutional Approval

Permission from school administration for activities that affect school operations.

People Resources

Research Team

Friends or classmates who will help conduct the research. List specific roles and responsibilities.

Participants

How many people will be in your study? How will you recruit them? What will they need to do?

Supervisors

Teachers, parents, or other adults who need to oversee activities, especially for safety or access reasons.

Experts

People with specialized knowledge who might advise on your project or help interpret results.



Timeline and Logistics



Planning Phase

Design research, create materials, get permissions, and prepare resources.



Data Collection

Conduct experiments, distribute surveys, perform interviews or observations.



Analysis

Organize data, perform calculations, identify patterns, and interpret findings.



Reporting

Write up results, create visuals, prepare presentation, and share conclusions.



Creating a Detailed Timeline



Schedule Data Collection

Plan exactly when you'll collect data, with specific dates and times.



Estimate Duration

Calculate how long each session or step will take to ensure it's feasible.



Include Waiting Periods

Account for necessary delays between steps, like plant growth or survey response time.



Set Internal Deadlines

Create target dates for completing each phase to stay on track throughout the project.



Flexibility and Troubleshooting

Anticipate Challenges

Think ahead about what might go wrong and how you'll handle it.

Have Backup Plans

Create alternative approaches for critical parts of your research.



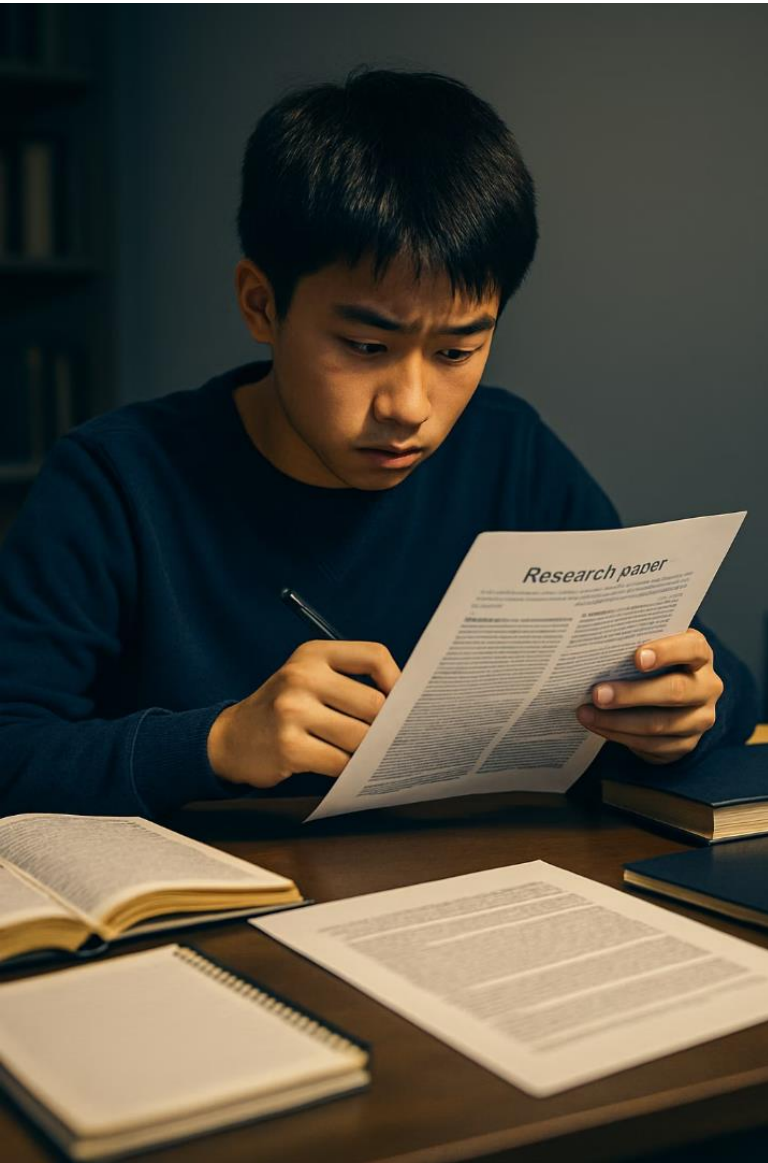
Be Ready to Adapt

Prepare to adjust your methods if you discover better approaches.

Document Changes

Record any modifications to your original plan and why you made them.

Literature Review



What is a Literature Review?

A literature review is a summary and analysis of what has already been written on a topic.

It helps you:

- Understand what's already known,
- See where there are gaps or debates,
- Avoid repeating others' work,
- And build a strong foundation for your own research.

How to Conduct a Lit Review

Step 1: Clarify Your Topic or Research Question

Step 2: Search for Reliable Sources

Step 3: Skim and Select

- Read abstracts first to decide if the source is relevant.
- Choose a mix of recent, relevant, and diverse perspectives.
- Aim for at least 5–10 solid sources (depending on assignment length).

Step 4: Take Notes

- Write down the main point of each source.
- Note how it relates to your topic, and any key quotes or data.
- Use color coding or a chart to organize themes or arguments.

How to Conduct a Lit Review

Step 5: Group by Themes or Ideas

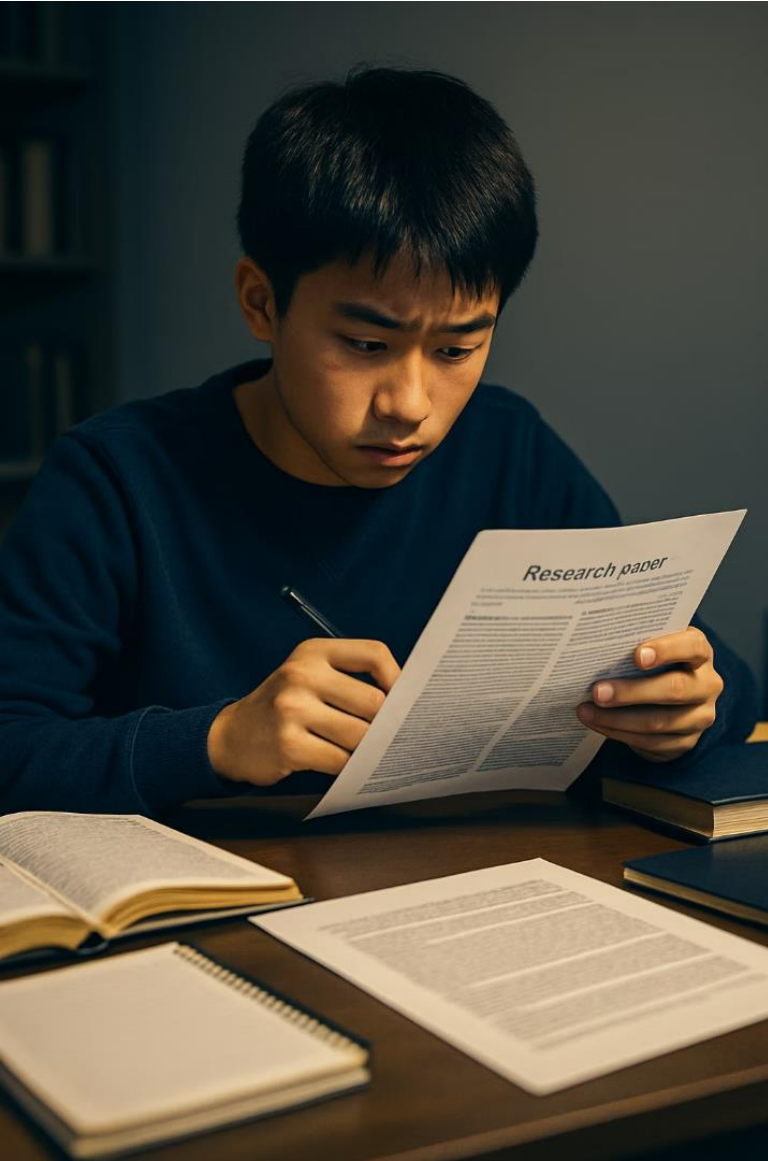
- Don't stop at summarizing each article one by one.
- Instead, group your notes by common themes, disagreements, or trends.
- Ask: What do these sources have in common? How do they differ?

Step 6: Start Writing

Step 7: Cite Everything

Literature Review Template

Short Version



- **Begin with a short introduction:**
 - What is your topic?
 - Why is it important?
- **Body:**
 - Share what other studies have found.
 - Organize the body by themes, not by author.
- **End with a summary:**
 - Clarify how your project connects to past research.

Importance of giving reference aka citing

What is Citing?

Citing your sources means giving credit to the original authors of the ideas, words, or data you use in your work.

We cite to:

- Avoid plagiarism (taking credit for someone else's work),
- Show where your information came from, and
- Help others find those sources if they want to learn more.

What to Cite? What not to Cite?

You should cite:

- Direct quotes (even short ones)
- Paraphrased ideas (putting someone else's idea into your own words)
- Statistics or data
- Images, charts, or graphics you didn't create

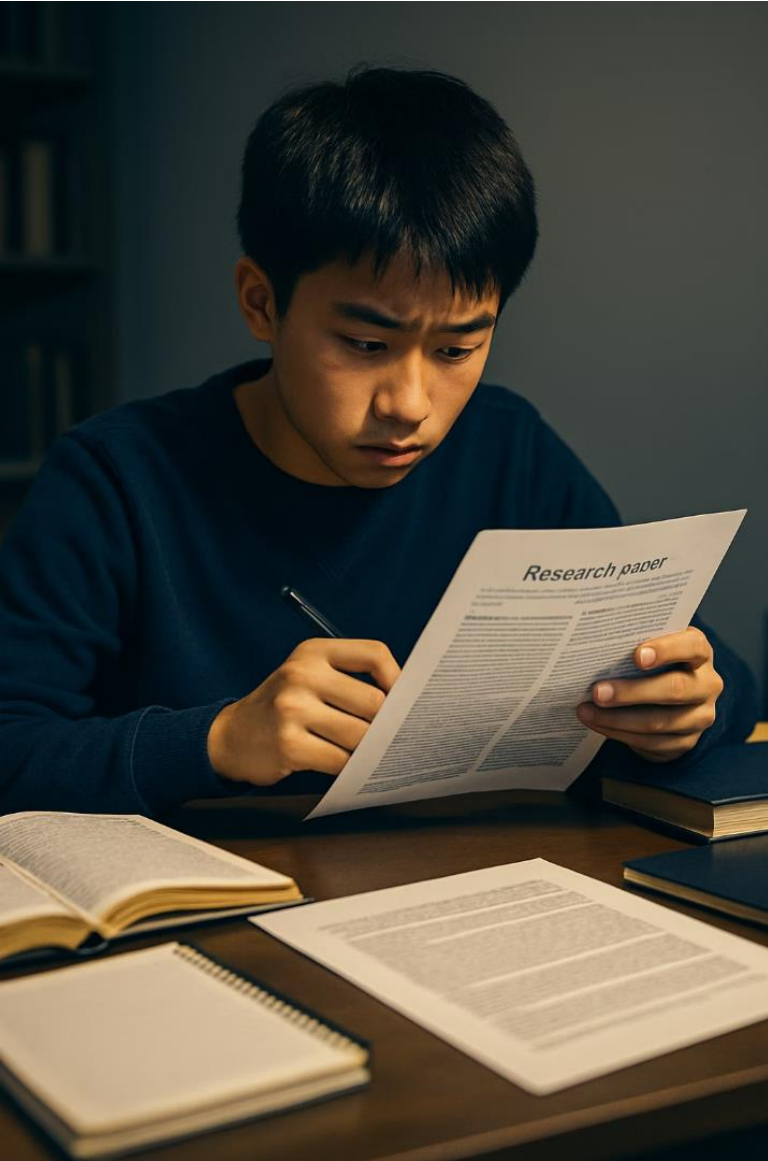
You do not need to cite:

- Common knowledge (e.g., "Water freezes at 0°C")
- Your own original thoughts

How to cite (more about this on Day 8):

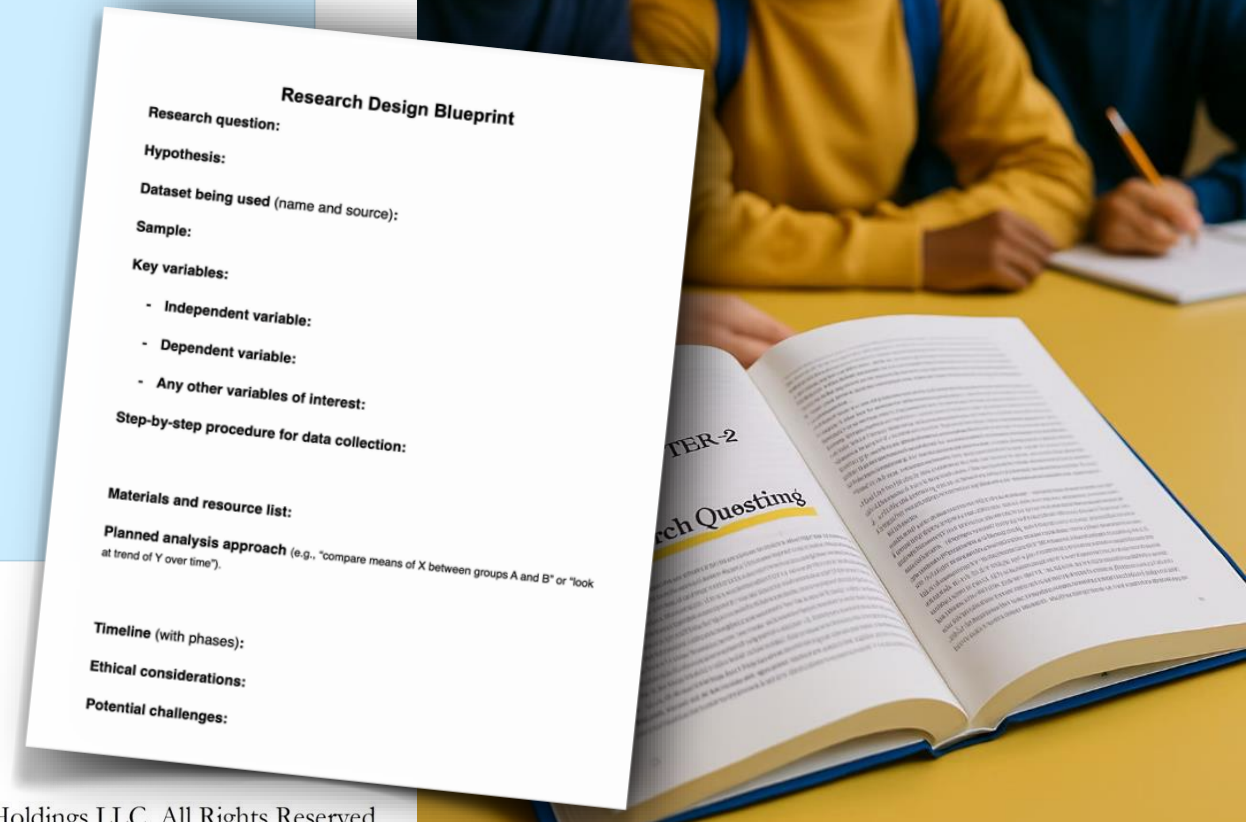
- Name of the first author & date - in the text
- Full info - on the references page

Hands-on Reading / Writing Time



Homework - Teamwork

- Finish Handout 4: Research Design Blueprint.
- Continue with your lit review. Make sure you have resources that speak to why your topic is important. **You would need this information for your upcoming presentation!** As always, keep track of your resources through “Source Tracker”. Share your notes with your team members.



EXTRA SLIDES FOR DAY 5

Common Research Challenges

Participation Issues

- Low survey response rates
- Participants dropping out
- Scheduling conflicts

Solution: Recruit more than needed, send reminders, offer incentives, be flexible with scheduling.

Measurement Problems

- Unclear results
- Equipment failures
- Inconsistent data

Solution: Test instruments beforehand, use multiple measures, document anomalies, increase sample size.

Resource Limitations

- Budget constraints
- Limited access
- Time shortages

Solution: Scale project appropriately, find alternatives, prioritize essential elements.

Ethical Considerations



Respect Privacy

Protect participants' identities and personal information. Use anonymous responses when possible.



Informed Consent

Ensure participants understand what they're agreeing to. Get permission from parents for minors.



Minimize Harm

Design research that doesn't cause physical or emotional distress to participants.



Get Approvals

Obtain necessary permissions from teachers, administrators, or other authorities.

